ECE 2305 – Introduction to C Programming
Tutorial 09
Boolean Operations

Write a C++ application to apply the following Boolean functions and display the results.

```
int a = 10;
int b;  // the variable b will take the values 5, 10 and 15
bool q;

q = a < b;
q = a <= b;
q = a == b;
q = a >= b;
q = a > b;

q = b > 5 && b > 15;
q = b > 5 && b < 15;
q = b < 5 && b > 15;
q = b < 5 && b < 15;

q = b > 5 || b > 15;
q = b > 5 || b < 15;
q = b < 5 || b > 15;
q = b < 5 || b < 15;
```

Write the program to display the value of **q**. Compare the results for **b = 5**, **b = 10** and **b = 15**.

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Input

```
// ECE 2305 - Tutorial 09
// Boolean Functions

#include<iostream>
using namespace std;

int main()
{
	int a = 10;
	int b = 5; // the variable will take the value 5, 10, or 15

	cout << "ECE 2305 - Tutorial 09" << endl;
	cout << "Boolean Operators" << endl;
	cout << endl;

	cout << "a = " << a << endl;
	cout << "b = " << b << endl;
	cout << endl;
	cout << "a < b = " << (a < b) << endl;
	cout << "a <= b = " << (a <= b) << endl;
	cout << "a == b = " << (a == b) << endl;
	cout << "a >= b = " << (a >= b) << endl;
	cout << "a > b = " << (a > b) << endl;
	cout << endl;
	cout << "b > 5 && b > 15 = " << (b > 5 && b > 15) << endl;
	cout << "b > 5 && b < 15 = " << (b > 5 && b < 15) << endl;
	cout << "b < 5 && b > 15 = " << (b < 5 && b > 15) << endl;
	cout << "b < 5 && b < 15 = " << (b < 5 && b < 15) << endl;
	cout << endl;
	cout << "b > 5 ||  b > 15 = " << (b > 5 || b > 15) << endl;
	cout << "b > 5 ||  b < 15 = " << (b > 5 || b < 15) << endl;
	cout << "b < 5 ||  b > 15 = " << (b < 5 || b > 15) << endl;
	cout << "b < 5 ||  b < 15 = " << (b < 5 || b < 15) << endl;
	cout << endl;
	system("pause");
	return(0);
}
```

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Output





